

Multi-Sensor Cough Analysis and State Classification with Machine Learning

by

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ABSTRACT

This project presents a wearable-sensor system to detect cough events and to understand the activity during coughing. The data come from multiple sensors, including a chest microphone, an ambient microphone, an accelerometer, and a stretch sensor. First, the signals are cleaned with basic filtering and divided into short time windows. Then, a small set of time and frequency features is extracted from each window.

Different machine learning models are trained and tested using file-based cross-validation to avoid using parts of the same recording in both training and testing. Finally, window predictions are combined to find cough time intervals, and the results are evaluated at the event level.

The best cough detection model achieves strong event-level performance with precision 0.925, recall 0.919, and F1-score 0.922 ($\text{IoU} \geq 0.2$). Activity classification between sitting and walking also performs well, with about 0.89 window-level accuracy and event-level F1-scores of 0.88 for sitting and 0.79 for walking. Although the dataset is limited and includes recordings from a single subject, the results show that using multiple wearable sensors with machine learning can detect coughs accurately and provide useful activity context.

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Chapter 1

INTRODUCTION

1.1 Background and Motivation

Cough is an important physiological signal that provides information about respiratory health. Continuous cough monitoring can support early detection of respiratory diseases, treatment follow-up, and long-term patient observation. However, reliable cough detection in daily-life conditions is challenging due to background noise, motion artifacts, and privacy concerns related to audio recordings[1].

Wearable sensing offers an alternative to conventional microphone-based systems[2]. Signals such as body-coupled audio and motion measurements are less affected by environmental noise and provide better robustness in real-world conditions. Nevertheless, these signals are often noisy, non-stationary, and highly dependent on user activity. As a result, both signal processing techniques and data-driven methods are required to achieve reliable performance.

Traditional digital signal processing (DSP) methods provide interpretable and low-complexity solutions. However, their performance may be limited when signal characteristics vary across conditions. Machine learning (ML) methods can model more complex patterns but require careful feature design, validation strategies, and post-processing to avoid overfitting and misleading performance estimates[3]. For these reasons, machine learning approaches offer greater flexibility than traditional DSP methods when signal characteristics vary.

This project investigates a multi-sensor, machine-learning-based cough detection and activity classification system using wearable audio and motion data. Signal preprocessing and feature extraction are applied prior to supervised classification, followed by event-level post-processing to improve robustness.

1.2 Problem Definition and Objectives

The main problem addressed in this study is the reliable detection of cough events and the classification of user activity from wearable sensor signals recorded in short-duration sessions. The system is required to operate under realistic conditions, where signals are affected by motion, environmental interference, and variability in cough characteristics.

The primary objectives of this project are:

- To develop an ML-based pipeline for cough detection using time- and frequency-domain features.
- To perform activity classification using wearable sensor data.
- To evaluate the methods using consistent window-level and event-level metrics.

Success is defined not only by high classification scores, but also by stable event-level detection performance and transparent evaluation procedures.

1.3 Contributions

The main contributions of this project can be summarized as follows:

- Development of a supervised machine learning pipeline for window-level cough detection using wearable audio.
- Extension of detected cough events to activity classification using motion features.

Chapter 2

RELATED WORK

Automatic cough detection has been widely studied using both audio-based and wearable sensing approaches[2]. Early systems mainly relied on digital signal processing techniques, where cough events were detected using energy-based segmentation, spectral analysis, and handcrafted temporal features. These methods provided interpretable pipelines but showed limited robustness under varying noise conditions and user activities[4].

With the availability of larger datasets and increased computational resources, machine learning methods have become dominant[3]. Audio-based systems using time–frequency representations and classifiers such as support vector machines, random forests, and neural networks have showed improved detection accuracy. More recently, deep learning models(CNN, RNN) operating on spectrograms or sequential features have been proposed to reduce the dependency on handcrafted features and improve generalization[5].

To address privacy concerns and environmental noise, wearable and multi-sensor systems have gained attention. Accelerometers, strain sensors, and chest-mounted transducers have been used either alone or in combination with audio signals[1]. These systems have shown that body-coupled signals can provide reliable cough detection while being less sensitive to ambient sound.

Chapter 3

DATASET AND SYSTEM

3.1 System Overview

This project uses a wearable multi-sensor system designed for cough detection and activity classification. The system records synchronized audio and motion-related signals, allowing both acoustic and body-coupled information to be analyzed jointly. The overall processing pipeline includes signal acquisition, preprocessing, feature extraction, and classification stages, which are shared by both DSP-based and machine learning-based methods.

Using multiple sensing modalities improves robustness against environmental noise and motion artifacts. In particular, combining body-coupled sensors with ambient audio enables more reliable cough detection under realistic conditions.

3.2 Sensor Configuration

Each recording contains four synchronized sensor channels stored as raw integer values obtained from analog-to-digital converters. The sensor channels are ordered as follows:

- **Pulmonary microphone:** captures internal chest sounds related to respiration and coughing,
- **Ambient microphone:** records environmental audio,
- **Stretch sensor:** measures chest wall expansion and contraction,
- **Accelerometer (Z-axis):** captures vertical body motion.

All sensors are sampled synchronously at a sampling rate of 4800 Hz. The accelerometer Z-axis was selected as it is sensitive to vertical movement and posture-related motion.

3.3 File Structure and Organization

The dataset is stored in comma-separated values (CSV) format. Each file corresponds to a single recording session with a duration of approximately 20 seconds. The files follow the naming convention:

$$\text{YYYYMMDD_ID_Activity_Condition.csv} \quad (3.1)$$

This naming scheme encodes the recording date, subject identifier, activity type, and recording condition. The file-based structure is preserved throughout all experiments. For model evaluation, data splits are performed at the file level to prevent temporal overlap between training and test data.

3.4 Dataset Composition

The dataset includes recordings collected under multiple activity types and environmental noise conditions. The activity and noise distributions are summarized in Table 3.1.

Table 3.1: Dataset distribution across activities and noise conditions

Activity	Clean	Cough	Music	Sneeze	Snooze	Door	Total
Sitting up	18	14	4	2	1	1	40
Walking	5	13	2	0	0	0	20
Running	0	0	1	0	0	0	1
Standing up	3	0	0	0	0	0	3

Given the insufficient sample size for *Running* and *Standing up*, the current work focuses on *Sitting up* and *Walking* to ensure stable and interpretable activity classification.

The presence of different noise sources, such as background music, external cough sounds, and door noises, allows evaluation of false positive behavior and noise robustness.

3.5 Raw Signal Characteristics

Figure 3.1 shows an example recording illustrating the raw sensor signals and the corresponding cough labels. Cough events appear as high-energy bursts in the pulmonary microphone signal and are accompanied by characteristic patterns in the stretch sensor and accelerometer channels. In contrast, non-cough sounds may be present in the ambient microphone without corresponding body-coupled responses.

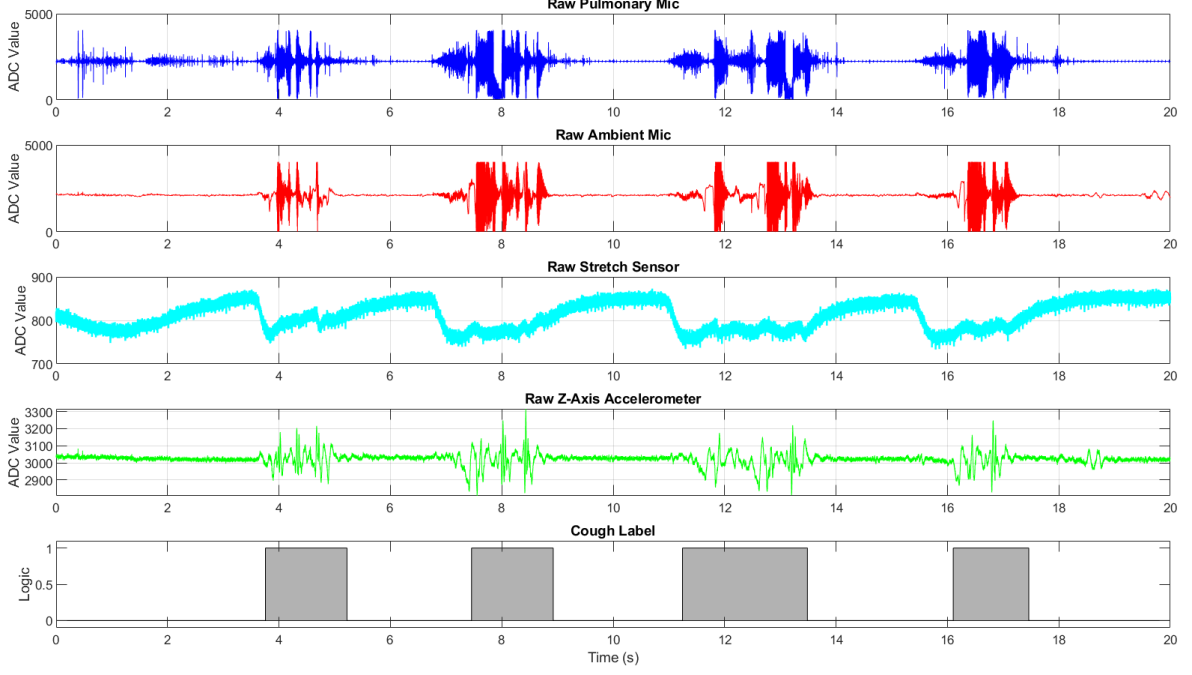


Figure 3.1: Example raw sensor signals and cough labels over a 20-second recording.

3.6 Ground Truth Encoding

The cough ground truth is embedded directly within the stretch sensor channel using bit-wise encoding. Specifically, the least significant bit of the stretch sensor value represents the cough label, while the remaining bits represent the sensor magnitude.

Let $x[n]$ denote the raw value of the third channel. The cough label $y[n]$ and the stretch sensor signal $s[n]$ are obtained as:

$$y[n] = x[n] \& 1, \quad (3.2)$$

$$s[n] = x[n] \gg 1. \quad (3.3)$$

Here, $y[n] = 1$ indicates the presence of a cough event, and $y[n] = 0$ indicates the absence of a cough. This encoding scheme allows both the signal and its annotation to be stored in a single integer-valued channel. All decoding operations are deterministic and causal, making the dataset suitable for real-time and offline analysis.

Chapter 4

METHODOLOGY

4.1 Preprocessing

Preprocessing is required before model creation because the raw sensor outputs contain hardware-related artifacts and mixed information that can mislead machine learning models. In this dataset, (i) one channel includes both signal and labels, (ii) the signals include baseline drift and (iii) high-frequency components close to the Nyquist limit are dominated by noise. If these issues are not handled, extracted features become unstable, the model may fit noise patterns, and event timing between channels can be distorted.

4.1.1 Band Limitation and Filtering Strategy

For the dataset being worked on, all recordings are sampled at $F_s = 4800$ Hz, so the Nyquist frequency is 2400 Hz. Digital filtering cannot remove aliasing if it already exists; therefore, any analog anti-alias filtering is assumed to be performed before sampling. The digital filters used here are applied to (i) remove artifacts and (ii) restrict analysis to physiologically meaningful frequency ranges.

4.1.2 Per-Channel Filtering

The following channel-specific filters are applied:

- **Pulmonary microphone (Col 1):** bandpass filtering in the range 60–2200 Hz. The high-pass component suppresses baseline drift and strong low-frequency interference. The low-pass component suppresses high-frequency noise near the Nyquist limit while preserving cough-relevant content.
- **Ambient microphone (Col 2):** the same bandpass filter (60–2200 Hz) is applied using identical coefficients. This enables direct and fair comparison between pulmonary and ambient audio features, because both channels are shaped by the same frequency response.

- **Stretch sensor (decoded Col 3):** low-pass filtering at 20 Hz. Most informative stretch dynamics are concentrated in the very low-frequency region, while components above approximately 15–20 Hz are largely noise dominated in this dataset.
- **Accelerometer Z-axis (Col 4):** low-pass filtering at 20 Hz, with DC preserved (no high-pass). Preserving DC is important because the gravity component supports posture and activity-related distinctions. The low-pass filter suppresses high-frequency jitter while retaining the dominant motion dynamics.

4.1.3 Edge Effects and Segment Handling

Zero-phase filtering introduces edge effects due to internal padding. To reduce this issue, each full 20-second recording is filtered as a single continuous signal, and segmentation for labels, events, and feature windows is performed after filtering. Under this approach, edge artifacts remain confined to the beginning and end of the recording and have limited impact on labeled events.

4.1.4 Example of Preprocessed Signals

An example of preprocessed signals is shown in Figure 4.1. The figure illustrates the effect of band-limiting and smoothing across channels, while maintaining alignment between cough labels and multi-sensor responses.

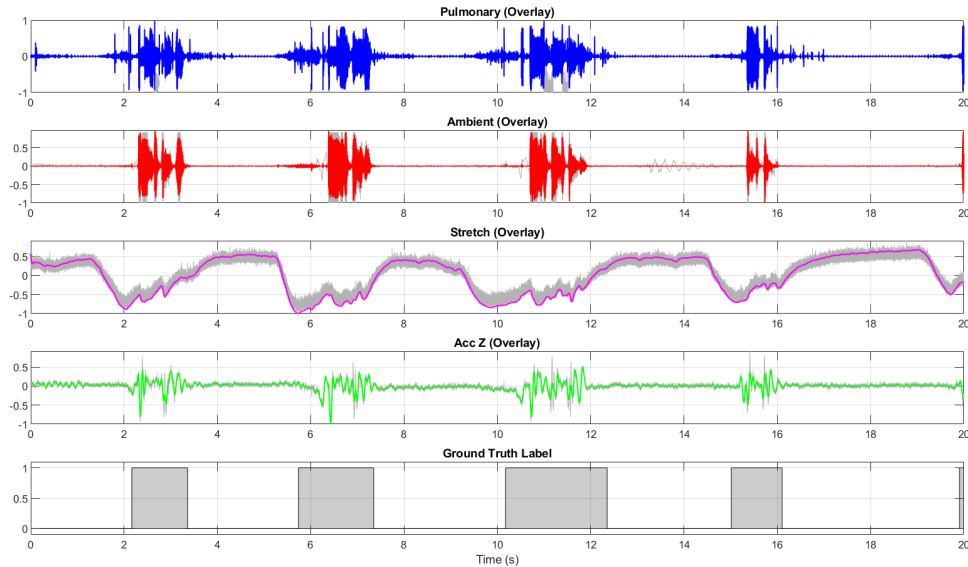


Figure 4.1: Example preprocessed sensor signals and cough labels over a 20-second recording.

4.2 Feature Extraction

Feature extraction aims to represent cough-related acoustic and mechanical patterns using a limited number of informative and robust descriptors. Although a large number of features can be derived from multi-sensor signals, using an excessive feature set increases model complexity and the risk of overfitting, especially when the dataset size is limited. For this reason, a compact feature set was selected based on direct signal observations and physical interpretation.

All features are computed on short-time frames of 200 ms with a hop size of 50 ms. Frame-level features are later aggregated for event-level classification when required.

4.2.1 Motivation for Feature Selection

Figure 4.2 illustrates an example recording with synchronized pulmonary microphone, ambient microphone, stretch sensor, accelerometer, and cough ground truth. Several consistent patterns motivated the chosen feature set:

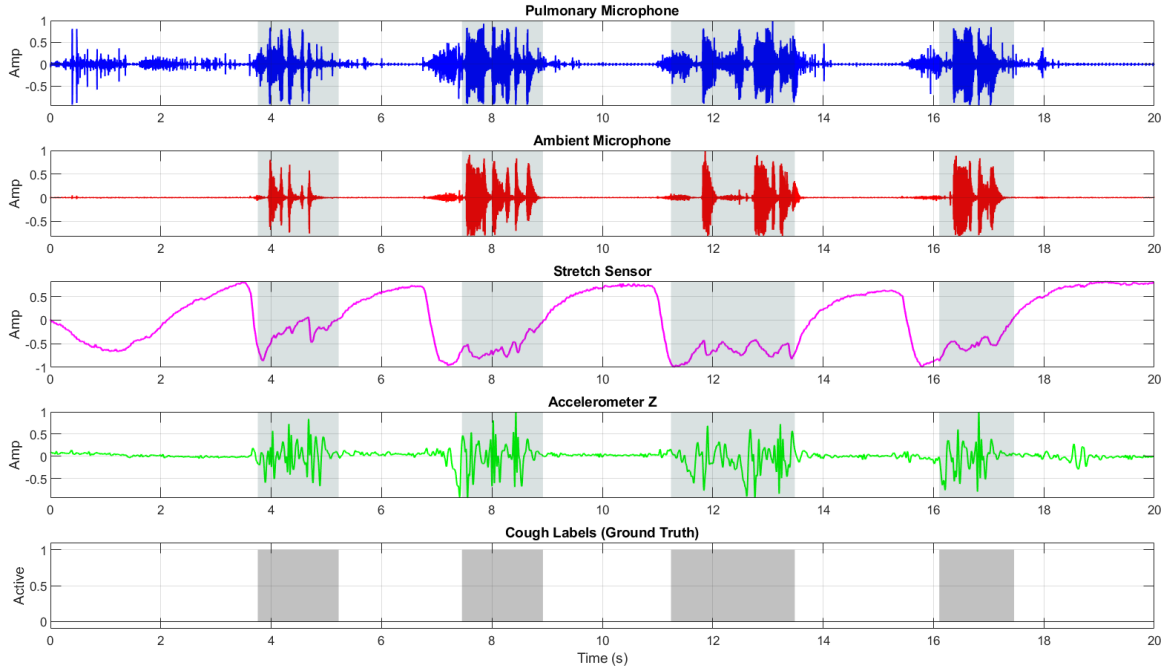


Figure 4.2: Example preprocessed sensor signals and cough labels

- Cough events produce short, high-energy bursts in the pulmonary microphone signal.
- During coughs, the accelerometer signal exhibits strong oscillations due to rapid chest vibration.
- The stretch sensor typically increases before the cough and reaches a pronounced minimum during the cough event.

- Background sounds may appear in the ambient microphone without corresponding body-coupled responses.
- Cough signals are impulsive and non-Gaussian, which motivates the use of higher-order statistics.

Based on these observations, the following time-domain and frequency-domain features were selected[15].

4.2.2 Time-Domain Features

Pulmonary Microphone RMS Energy

Cough events are characterized by sudden energy bursts in the pulmonary microphone. The root mean square (RMS) energy of a frame $x[n]$ is computed as

$$\text{RMS}_p = \sqrt{\frac{1}{N} \sum_{n=0}^{N-1} x^2[n]}. \quad (4.1)$$

The logarithm of this value is used to reduce dynamic range and improve numerical stability.

Accelerometer Variance

Rapid chest motion during coughing causes strong oscillations in the accelerometer signal. This behavior is captured using the variance:

$$\sigma_{\text{acc}}^2 = \frac{1}{M} \sum_{m=0}^{M-1} (a[m] - \bar{a})^2. \quad (4.2)$$

This feature is also highly informative for activity classification.

Stretch Signal Minimum

The stretch sensor shows a characteristic drop during cough events. The minimum stretch value within a frame is used:

$$s_{\min} = \min_m(s[m]). \quad (4.3)$$

This feature captures the mechanical compression associated with coughing.

Stretch Signal Sharpness

To quantify how abruptly the stretch signal changes, sharpness is computed as the mean absolute first derivative:

$$S_{\text{sharp}} = \frac{1}{M-1} \sum_{m=1}^{M-1} |s[m] - s[m-1]|. \quad (4.4)$$

Pulmonary-to-Ambient Energy Ratio

To separate body-generated coughs from environmental sounds, the ratio between pulmonary and ambient microphone energies is used:

$$R = \frac{\text{RMS}_p}{\text{RMS}_a + \epsilon}, \quad (4.5)$$

where ϵ is a small constant. The logarithm of this ratio is included as a feature.

Event Duration

For event-level classification, the duration of detected candidate segments is included as an additional feature. Cough events typically have short and consistent durations compared to non-cough events.

4.2.3 Frequency-Domain and Statistical Features

Spectral Centroid

The spectral centroid describes the center of mass of the pulmonary power spectrum[8]:

$$C = \frac{\sum_k f_k P[k]}{\sum_k P[k]}. \quad (4.6)$$

Cough sounds generally exhibit higher centroid values due to their broadband nature.

Audio Kurtosis

Kurtosis is used to quantify impulsiveness in the pulmonary microphone signal[7]:

$$\kappa_{\text{audio}} = \frac{\mathbb{E}[(x - \mu)^4]}{\sigma^4}. \quad (4.7)$$

High kurtosis values indicate sharp, transient events such as coughs.

Stretch Signal Kurtosis

Similarly, kurtosis is computed for the stretch signal[7]:

$$\kappa_{\text{stretch}} = \frac{\mathbb{E}[(s - \mu_s)^4]}{\sigma_s^4}. \quad (4.8)$$

This feature captures sudden mechanical impulses during coughing.

4.2.4 Role of Accelerometer Z-Axis in Activity Classification

The accelerometer Z-axis is an important signal for activity classification because it reflects vertical body motion. Figure 4.3 compares sitting and walking recordings.

During sitting, the accelerometer signal is mostly stable and shows small fluctuations. In contrast, during walking, the signal becomes strongly oscillatory due to repeated body movement. This difference is clearly visible in the amplitude and variability of the signal.

Because of this behavior, features such as accelerometer variance provide strong information for separating sitting and walking activities.

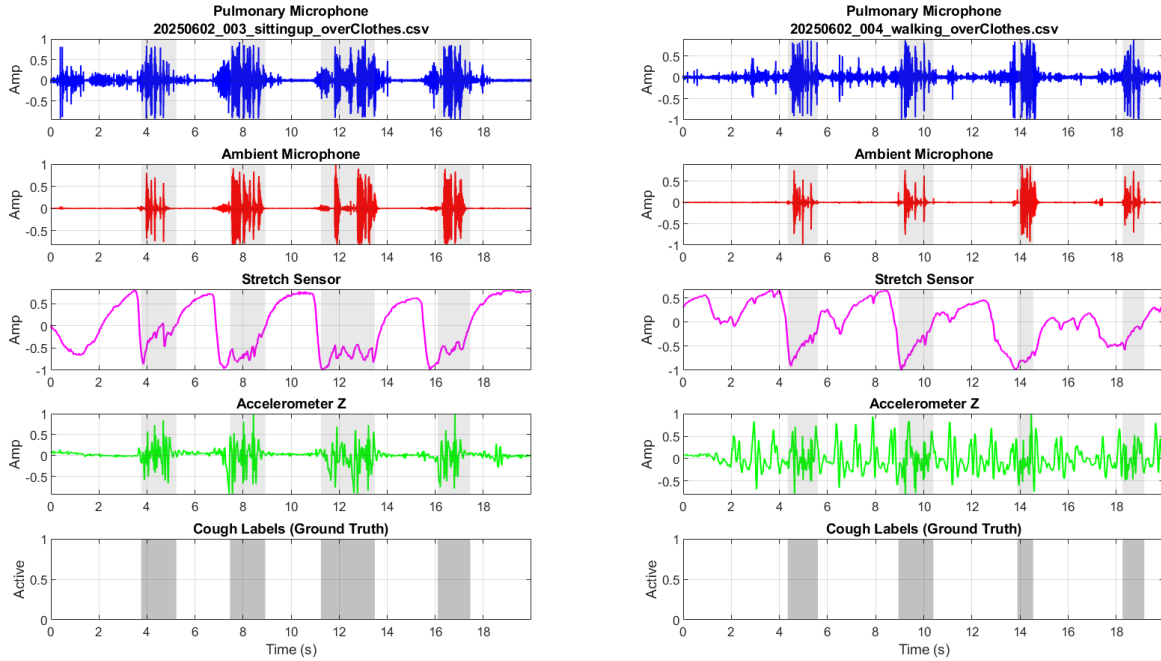


Figure 4.3: Multi-sensor signals for sitting (left) and walking (right)

4.2.5 Feature Aggregation and Robustness

Frame-based features are aggregated over detected candidate intervals. The final feature set was intentionally kept small to improve generalization and reduce sensitivity to noise and subject-specific variations.

4.2.6 Illustrative Spectrogram Example

Figure 4.4 shows an example spectrogram of the pulmonary microphone signal during a cough event. The broadband energy distribution and rapid temporal transitions motivate the use of spectral centroid, flatness, and flux features[8].

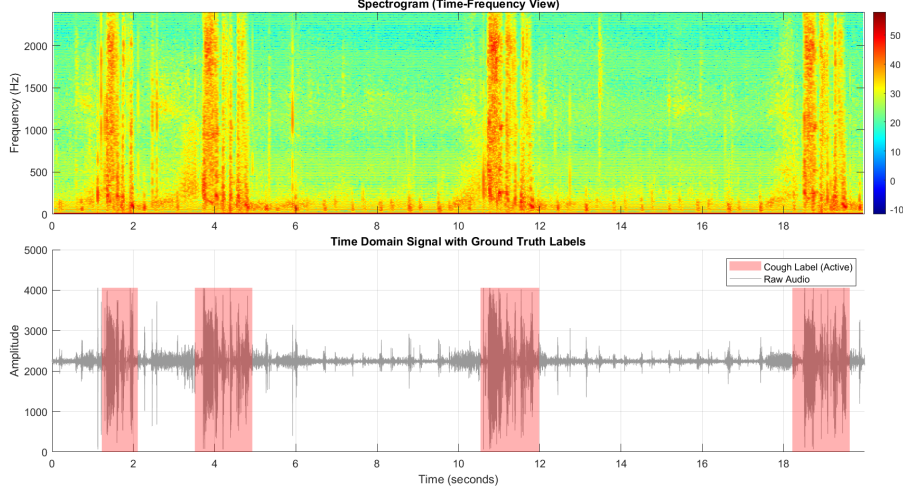


Figure 4.4: Time–frequency Spectrogram plot of a pulmonary microphone signal

While spectral flatness and spectral flux were initially considered as potential features based on spectrogram characteristics, the final feature set was limited to spectral centroid and higher-order statistics, as these provided sufficient discrimination with lower dimensionality.

4.3 Modeling

This study adopts a machine learning-based approach for both cough detection and activity classification. The modeling pipeline is designed as a two-stage system: first, cough events are detected from continuous multi-sensor recordings, and second, the activity context associated with these events is classified. Both tasks are formulated as supervised classification problems and are trained using the same underlying dataset and feature extraction framework.

4.3.1 Overview of the Two-Model Approach

Two separate models are developed:

- **Cough Detection Model:** a binary classifier that predicts whether a short-time window contains a cough event.
- **Activity Classification Model:** a binary classifier that predicts the activity state (sitting or walking) associated with a window or a detected cough event.

This separation allows each model to focus on a distinct objective and feature relevance, while sharing common preprocessing and feature extraction steps.

4.3.2 Dataset Preparation and Windowing

The dataset used for modeling consists of 60 recording files, with 40 files labeled as sitting and 20 files labeled as walking. Each file has a fixed duration of 20 seconds and is processed independently.

Continuous signals are segmented into overlapping windows with a duration of 0.2 seconds and a hop size of 0.05 seconds. This configuration provides a balance between temporal resolution and statistical stability, and ensures that short cough events are captured within at least one window. Using this windowing strategy, a total of 23,820 windows are generated across all files, with an overall cough-positive window rate of approximately 15.5%.

Window-level cough labels are assigned using an overlap-based rule with ground-truth cough intervals. A window is labeled as positive if its normalized temporal overlap with any ground-truth cough event exceeds a predefined threshold.

4.3.3 Frequency Analysis Parameters

Frequency-domain features are computed using the short-time Fourier transform (STFT). For audio signals, an FFT size of $N_{\text{FFT}} = 1024$ is used, which provides sufficient frequency resolution to capture the broadband characteristics of cough sounds while remaining computationally efficient at a sampling rate of 4800 Hz.

For motion-related signals, a smaller FFT size of $N_{\text{FFT}} = 256$ is used after downsampling to 100 Hz. This choice is motivated by the lower frequency content of motion signals and avoids unnecessary spectral resolution in frequency bands dominated by noise.

4.3.4 Feature Set

Each window is represented by a compact feature vector derived from time-domain, frequency-domain, and statistical descriptors. The final feature set used for modeling includes:

- Logarithmic RMS energy of the pulmonary microphone,
- Logarithmic ratio between pulmonary and ambient microphone energies,
- Spectral centroid of the pulmonary microphone signal,
- Variance of the accelerometer Z-axis,
- Minimum value of the stretch sensor signal,

- Sharpness of the stretch sensor signal,
- Kurtosis of the pulmonary microphone signal,
- Kurtosis of the stretch sensor signal.

For event-level activity classification, the duration of detected cough events is included as an additional feature.

4.3.5 Model Architectures

Four different classifiers are evaluated for the cough detection task:

- Logistic Regression with L2 regularization,
- Support Vector Machine with radial basis function kernel,
- Random Forest classifier,
- Extreme Gradient Boosting (XGBoost).

Standardization is applied where required using a `StandardScaler`. Class imbalance is handled using class-weighted loss functions or scale adjustment parameters, depending on the model.

The same set of classifiers is considered for activity classification, with emphasis on tree-based models for event-level analysis due to their robustness to feature scaling and non-linear relationships.

4.4 Validation Strategy

4.4.1 Single-Subject Consideration

All recordings in this study were collected from a single subject. As a result, inter-subject generalization cannot be evaluated at this stage. To avoid performance results that appear better than they really are because of similarities over time, the training and testing data are carefully kept separate by file.

4.4.2 Group-Based Cross-Validation

Windowed samples extracted from the same recording file are highly correlated. Therefore, standard random cross-validation would lead to data leakage, where nearly identical windows appear in both training and testing sets.

To prevent this, **GroupKFold cross-validation** is used, where each recording file is treated as one group. All windows originating from the same file are assigned to the same fold. In this study, $N_{\text{splits}} = 5$ folds are used[6].

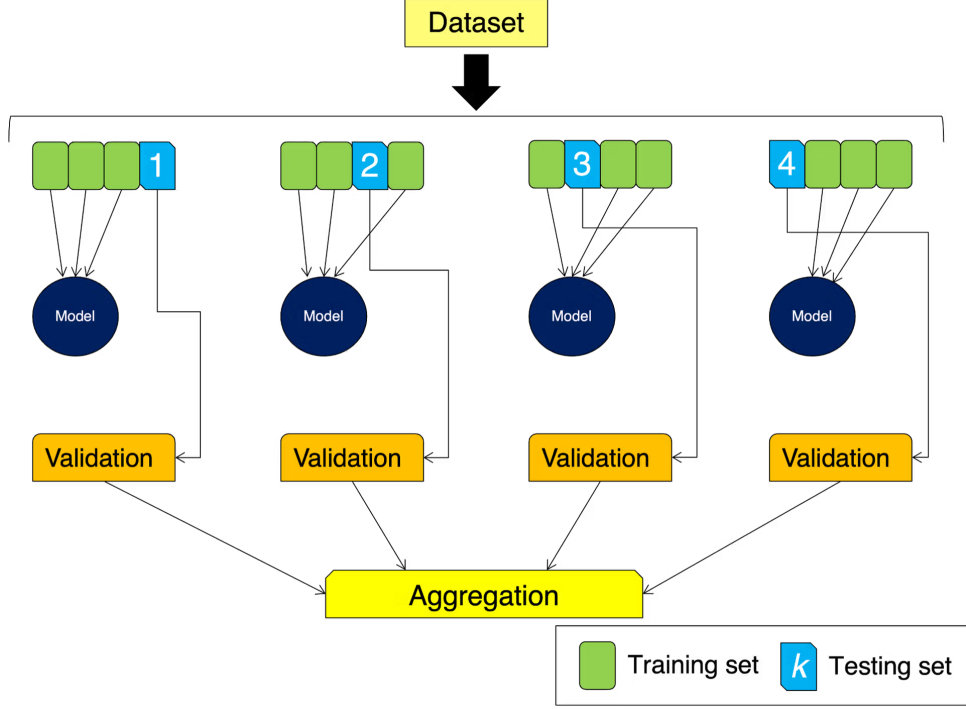


Figure 4.5: GroupKFold cross-validation strategy

This strategy ensures that model performance reflects generalization to unseen recordings rather than memorization of temporal patterns.

4.4.3 Class Imbalance Handling

Due to the sparsity of cough events, only approximately 15–18% of window-level samples are labeled as cough. To address this imbalance, class-weighted loss functions are used during model training. Specifically, positive cough samples are assigned higher weights, ensuring that minority-class errors are penalized more strongly during optimization.

4.4.4 Window-Level Evaluation

At the window level, each short-time segment is classified independently as cough or non-cough. Model performance is evaluated using out-of-fold predictions and standard classification metrics that are suitable for imbalanced datasets.

Let TP , FP , and FN [9] denote the number of true positives, false positives, and false negatives, respectively. The following metrics are used:

- **Precision**, which measures how many predicted cough windows are true coughs:

$$\text{Precision} = \frac{TP}{TP + FP}$$

- **Recall**, which measures how many true cough windows are detected:

$$\text{Recall} = \frac{TP}{TP + FN}$$

- **F1-score**, defined as the harmonic mean of precision and recall:

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

- **ROC-AUC**, which evaluates class separability across all decision thresholds.
- **PR-AUC**, which summarizes the precision–recall trade-off and is particularly informative when the positive class is rare:

$$\text{PR-AUC} = \int_0^1 \text{Precision}(\text{Recall}) d(\text{Recall})$$

Confusion matrices and classification reports are used to analyze error patterns. Window-level evaluation provides insight into local classification performance but it does not directly reflect event-level detection quality which is our desired result.

4.4.5 Event-Level Evaluation

For practical cough monitoring, event-level detection is more meaningful than window-level classification. Therefore, window-level probabilities are post-processed to form cough event candidates.

Predicted cough events are matched with ground-truth cough events using an intersection-over-union (IoU) criterion[10]. For a predicted event E_p and a ground-truth event E_g , the IoU is defined as:

$$\text{IoU}(E_p, E_g) = \frac{|E_p \cap E_g|}{|E_p \cup E_g|}. \quad (4.9)$$

A predicted event is counted as a true positive if its IoU with any ground-truth event exceeds a predefined threshold:

$$\text{IoU} \geq 0.20.$$

Unmatched predicted events are counted as false positives, and unmatched ground-truth events are counted as false negatives.

Table 4.1 summarizes the differences between window-level and event-level evaluation.

Table 4.1: Comparison of window-level and event-level evaluation

Aspect	Window-Level	Event-Level
Unit of analysis	Fixed window	Time interval
Label source	Overlap rule	Ground-truth events
Sensitivity to noise	High	Lower
Primary metric	F1, AUC	Event precision/recall

Chapter 5

IMPLEMENTATION AND RESULTS

This chapter presents the experimental evaluation of the machine learning models used for cough detection. All results are obtained using out-of-fold[11] predictions generated by GroupKFold cross-validation, ensuring that no data leakage occurs between training and testing sets.

5.1 Evaluation Metrics

Because cough windows represent only approximately 15–18% of all windows, accuracy is not a reliable performance measure. Instead, evaluation focuses on precision, recall, F1-score, ROC-AUC, and PR-AUC, as defined in Section 4.4.4.

Among these metrics, PR-AUC is considered the most informative for window-level cough detection. It directly reflects the trade-off between sensitivity and false alarms under class imbalance and allows fair comparison between models across different decision thresholds.

5.2 Cough Detection Results

Window Level

Four classifiers were trained and evaluated for window-level cough detection: Logistic Regression, SVM with RBF kernel, Random Forest and XGBoost. Table 5.1 summarizes their performance.

A total of 23,820 window-level samples are used for modeling, each represented by an 8-dimensional feature vector. The activity labels are imbalanced, with 15,880 windows corresponding to sitting and 7,940 windows corresponding to walking. This imbalance is handled during training using class-weighted loss functions.

Table 5.1: Window-level cough detection performance (out-of-fold)

Model	PR-AUC	ROC-AUC	F1	Precision	Recall
Random Forest (300)	0.903	0.964	0.827	0.910	0.757
XGBoost	0.900	0.965	0.797	0.733	0.873
SVM (RBF)	0.863	0.950	0.802	0.836	0.770
Logistic Regression	0.772	0.933	0.672	0.549	0.865

The results show that Random Forest and XGBoost achieve the highest PR-AUC values, indicating better discrimination of cough windows under class imbalance. Tree-based models are able to capture non-linear interactions between acoustic and motion features, which is critical for reliable cough detection.

5.2.1 Confusion Matrix Analysis

Confusion matrices provide insight into the error characteristics of each model. Figure 5.1 shows the confusion matrices for all evaluated classifiers.

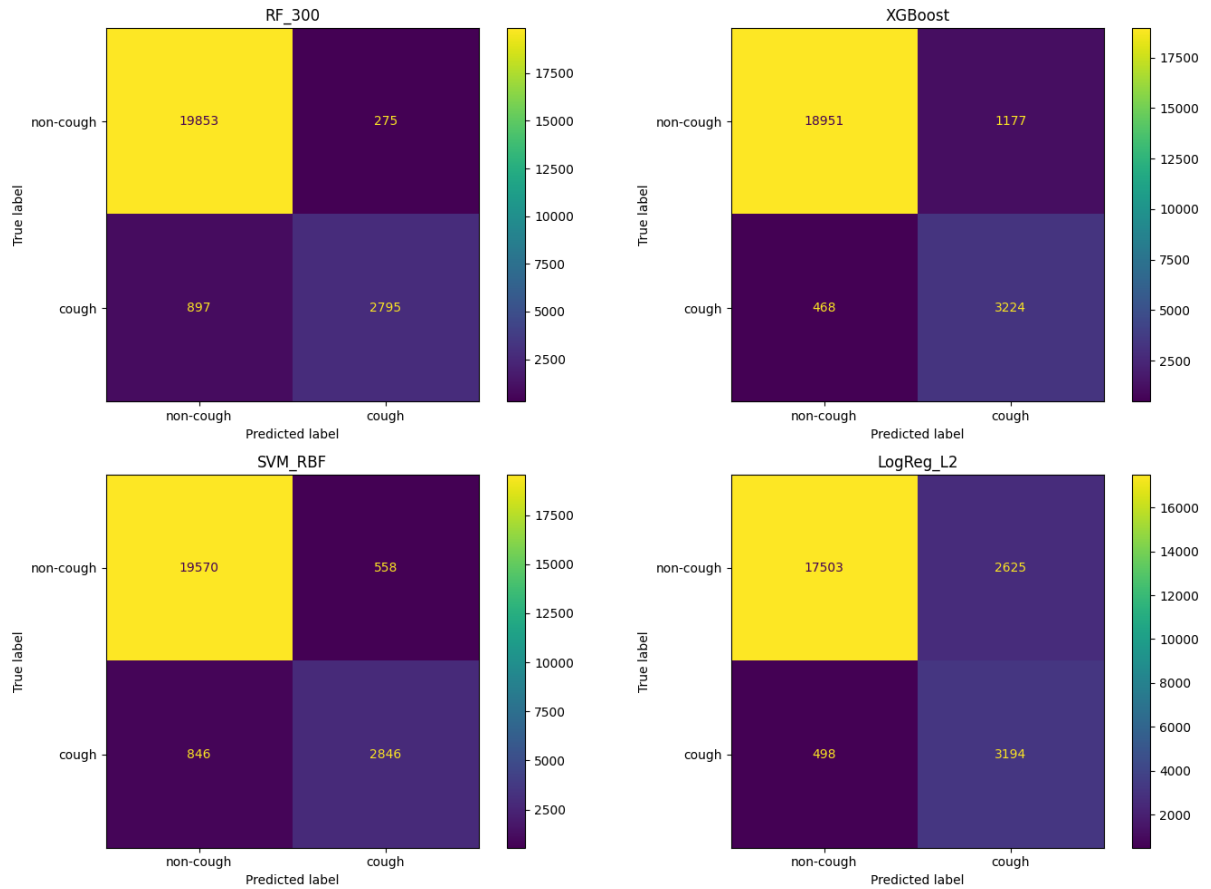


Figure 5.1: Window-level confusion matrices for cough detection models

Random Forest achieves high precision with relatively few false positives, while XG-

Boost provides higher recall by detecting a larger fraction of cough windows at the cost of increased false positives. This trade-off is relevant for downstream event detection, where missed coughs and spurious detections have different practical implications.

5.2.2 Precision–Recall Curve Analysis

Precision–recall curves illustrate model behavior across different decision thresholds. Figure 5.2 shows the PR curves computed from out-of-fold probabilities.

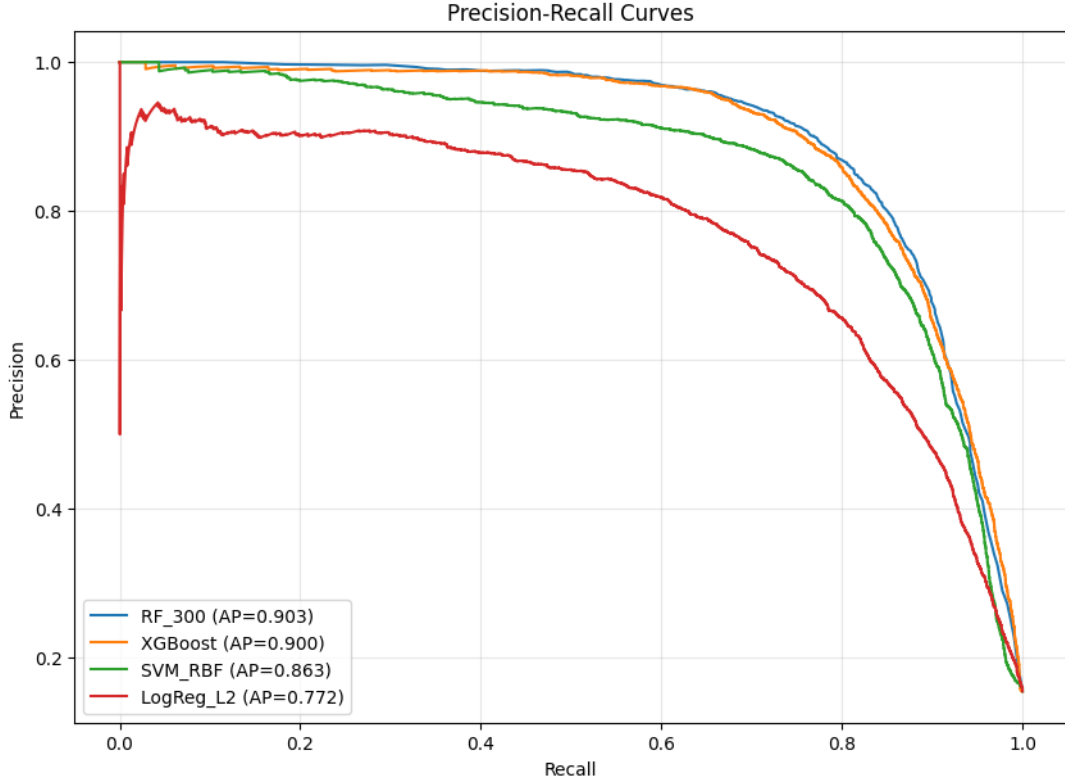


Figure 5.2: Precision–recall curves for window-level cough detection models.

Random Forest and XGBoost consistently dominate the PR space, maintaining high precision over a wide range of recall values.

Logistic Regression shows high recall but low precision which suggests that linear decision boundaries are insufficient for separating cough and non-cough windows.

Event-Level

Window-level predictions are useful for model comparison, but cough monitoring is naturally an event detection problem. For this reason, window-level probabilities are converted into continuous cough intervals using the post-processing procedure described in Section 4.4.5. The resulting predicted events are then compared with the ground-truth cough intervals using the IoU matching rule.

5.2.3 Metrics and Relevance

Event-level evaluation focuses on whether a cough event was detected with sufficient overlap, instead of whether every window was classified correctly. This is important because a single cough event can span multiple windows, and small boundary errors may not be meaningful in practice.

A predicted event is counted as a true positive if it matches any ground-truth event with $\text{IoU} \geq 0.20$. Unmatched predicted events are counted as false positives and unmatched ground-truth events are counted as false negatives.

5.2.4 Model Performance Evaluation

Event-level performance was then evaluated using the XGBoost model after post-processing. The resulting counts and metrics are reported in Table 5.2.

Table 5.2: Event-level cough detection results ($\text{IoU} \geq 0.2$)

Model	TP	FP	FN	Precision	Recall	F1
XGBoost	147	12	13	0.925	0.919	0.922

These results and figure 5.3 indicate that most cough events are detected with sufficient overlap, while the number of false alarms remains low. Compared to window-level evaluation, event-level metrics provide a clearer view of practical detection quality.

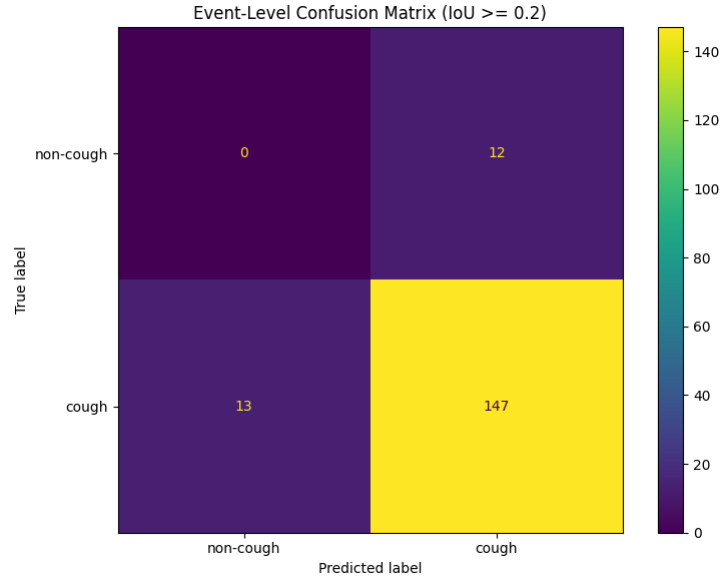


Figure 5.3: Event-level confusion matrix for XGBoost

5.3 Activity Classification Results

This section presents the results of activity classification for sitting and walking. Both window-level and event-level evaluations are reported to show how performance changes when predictions are aggregated over detected cough events.

5.3.1 Window-Level Activity Classification

For window-level activity classification, all 23,820 windows are used. Each window is represented by an 8-dimensional feature vector and inherits the activity label of its source recording. The class distribution is imbalanced, with 15,880 sitting windows and 7,940 walking windows. A window was labeled as a cough if at least 20% of its duration overlapped with a ground-truth cough interval ($\tau_{\text{overlap}} = 0.2$).

Two classifiers are evaluated: Random Forest and XGBoost. GroupKFold cross-validation with five folds is used to ensure that windows from the same recording do not appear in both training and testing sets.

Table 5.3 summarizes the window-level performance.

Table 5.3: Window-level activity classification performance

Model	Accuracy	F1-score	Precision	Recall
Random Forest	0.887	0.824	0.861	0.790
XGBoost	0.886	0.822	0.857	0.790

Random Forest and XGBoost achieve a decent accuracy and F1-score indicating that non-linear models better capture the relationship between motion-based features and activity state.

Figure 5.4 shows the confusion matrix for the Random Forest model.

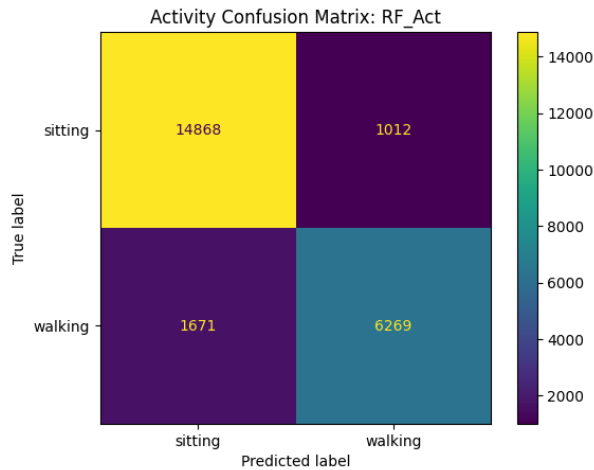


Figure 5.4: Window-level activity confusion matrix for Random Forest

5.3.2 Event-Level Activity Classification

Event-level activity classification is performed using detected cough events as units of analysis. For each cough event, features are aggregated across all windows belonging to that event, and event duration is added as an additional feature. This results in a dataset of 162 cough events, with 102 events labeled as sitting and 60 as walking.

A Random Forest classifier is trained and evaluated on this event-level dataset. Table 5.4 reports the classification performance.

Table 5.4: Event-level activity classification report

Class	Precision	Recall	F1-score	Support
Sitting	0.88	0.87	0.88	102
Walking	0.79	0.80	0.79	60

Figure 5.5 shows the event-level confusion matrix.

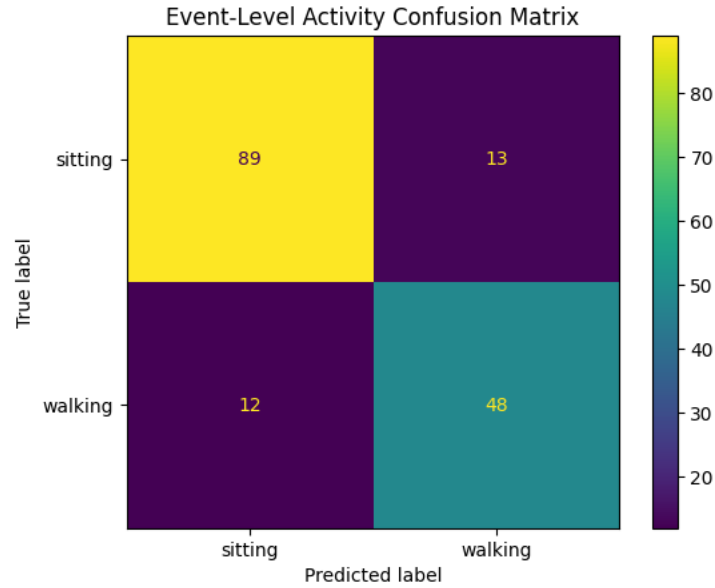


Figure 5.5: Event-level activity confusion matrix for Random Forest

5.3.3 Comparison and Discussion

Compared to window-level classification, event-level activity classification provides a more stable and interpretable result. By aggregating features over complete cough events, short-term noise and window boundary effects are reduced. Although the event-level dataset is smaller, the results indicate that activity context can be reliably inferred from cough-related motion patterns especially when using accelerometer-based features.

5.4 Feature Importance and Interaction Analysis

To better understand the behavior of the trained models, feature importance and feature interaction analyses are performed for both cough detection and activity classification. Figure 5.6 shows the feature importance scores, while Figure 5.7 presents the pairwise feature correlation heatmaps.

Feature Importance

For cough detection, the most influential features appeared to be the pulmonary microphone RMS energy, accelerometer variance, and stretch signal minimum. These features correspond directly to observable signal patterns, such as sudden energy bursts in audio signals, strong body vibrations, and rapid chest compression during coughing. This confirms that the model relies on physically meaningful and interpretable cues rather than spurious patterns.

For activity classification, the results indicate that the accelerometer variance is the dominant feature. Other features contribute very little to the decision process, which is expected because sitting and walking mainly differ in their motion characteristics. The limited contribution of non-motion features suggests that the model is not overfitting to irrelevant information. It is noted that the importance of other features may increase if additional activities, such as running, are included in the dataset.

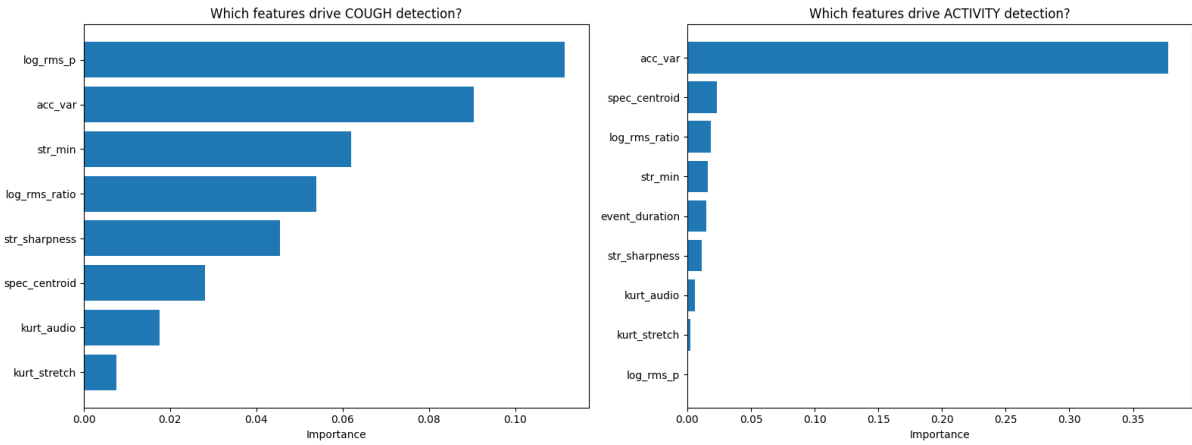


Figure 5.6: Feature importance for cough detection (left) and activity classification (right).

Feature Interaction

Feature interaction is examined using Pearson correlation heatmaps. Figure 5.7 shows that no pair of features exhibits a very strong correlation. This indicates that the selected features provide complementary information and that redundant features are limited.

The absence of strong correlations supports the robustness of the feature set and suggests that unnecessary or highly dependent features were not included during feature selection.

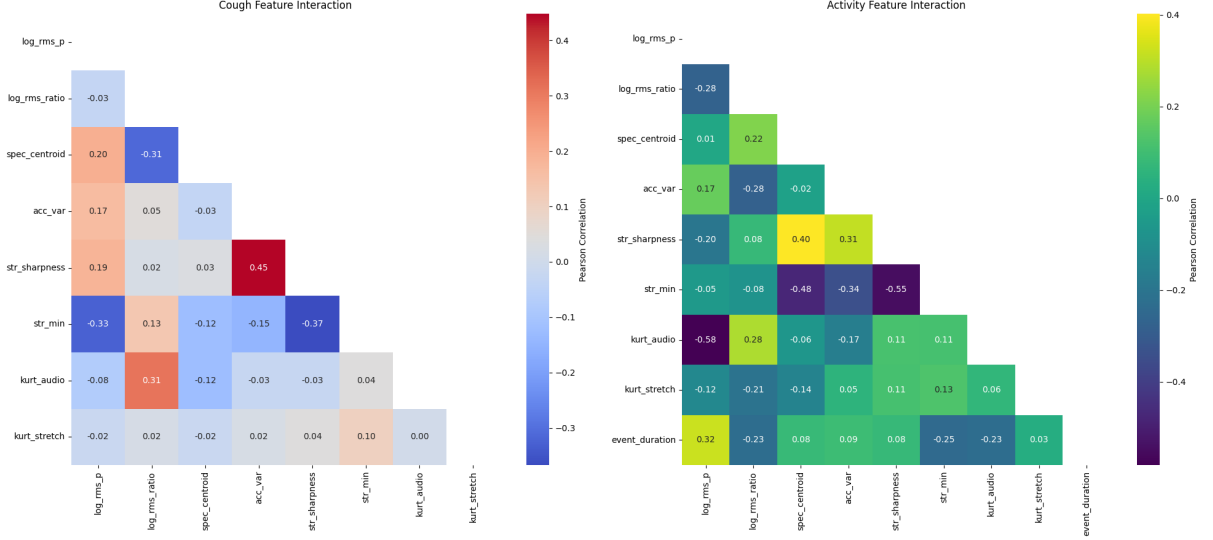


Figure 5.7: Pairwise feature correlation heatmaps for cough detection (left) and activity classification (right).

5.4.1 End-to-End System Demonstration

An end-to-end inference example is performed to demonstrate the complete operation of the system. A single recording is given as input, and cough events and their activity context are estimated without using ground-truth information.

The system first computes window-level cough probabilities, then converts these probabilities into cough events using post-processing. For each detected event, activity classification is performed using sensor features.

Figure 5.8 shows an example output. Clear cough events are detected at time regions with strong acoustic and motion responses. The system correctly labels the activity context of these events as sitting, demonstrating that cough detection and activity classification can be performed jointly in a single pipeline.

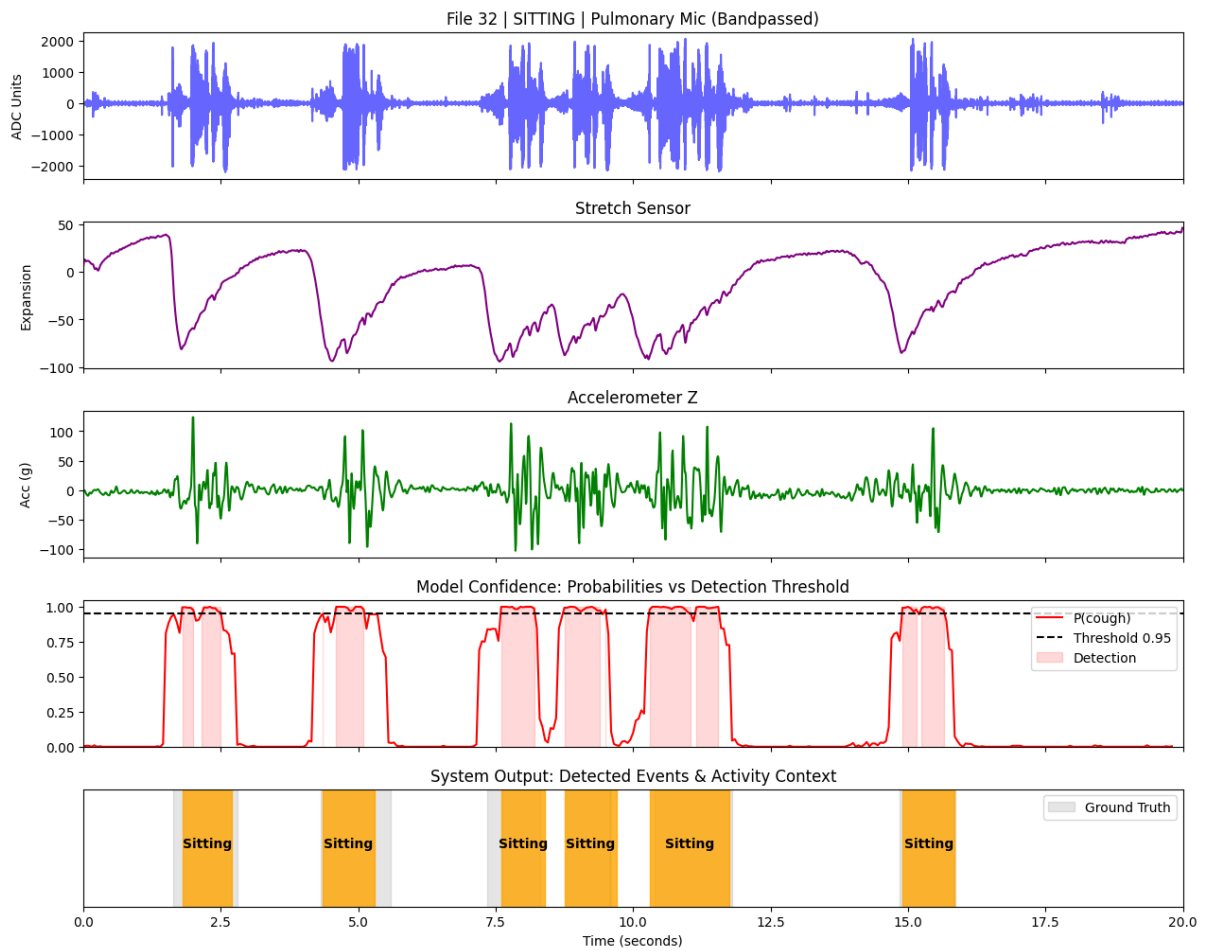


Figure 5.8: End-to-end system output showing detected cough events and their activity context

Chapter 6

CONCLUSION

6.1 Results

In this project, a multi-sensor machine learning system for cough detection and activity classification was developed and evaluated. The system successfully detects cough events and estimates their activity context using synchronized audio and motion signals.

At the window level, Random Forest and XGBoost models achieved the best performance for cough detection with high PR-AUC values under class imbalance. After post-processing, event-level evaluation showed high precision and recall which indicates that most cough events were detected with low false alarm rates.

Activity classification results demonstrated that sitting and walking can be reliably separated using motion-based features. In particular, accelerometer variance was found to be the most informative feature for activity recognition. Event-level activity classification further improved result stability by reducing window-level noise.

Overall, the results confirm that combining body-coupled sensors with machine learning provides an effective approach for cough monitoring and activity-aware analysis.

6.2 Discussion and Future Work

Although the proposed system shows promising results, several limitations and possible extensions are identified.

- **Dataset size and diversity:** The current study uses 60 recordings of 20 seconds collected from a single subject. This limits the ability to evaluate generalization across different individuals. Expanding the dataset with more subjects, longer recordings, and additional sessions is the primary future goal.
- **Deep learning models:** Deep learning methods using time-frequency representations such as spectrograms will be investigated. These models may learn more

complex patterns directly from the signals and improve detection performance.

- **DSP-based approaches:** While this work focuses on machine learning methods, purely DSP-based cough detection techniques could be explored as alternative or complementary solutions. Such approaches may provide simpler and more interpretable baselines.
- **Near real-time operation:** The current system is designed for offline analysis. Future work could focus on adapting the pipeline for near real-time or real-time operation by using causal filtering and low-latency processing.
- **Model optimization:** Machine learning performance could be further improved through systematic hyperparameter tuning and by testing additional classifiers and ensemble strategies.

6.3 Realistic Constraints

In this project, several realistic constraints related to data availability, cost, standards, and potential impact were considered. These constraints influence both the design choices and the interpretation of the results.

6.3.1 Social, Environmental and Economic Impact

The primary social impact of this project lies in its potential use for health monitoring applications. A reliable cough detection system may support long-term respiratory monitoring and provide useful information for clinicians and patients. However, the current system is a prototype and is not intended for clinical diagnosis.

From an environmental perspective, the project has minimal impact. It relies on low-power wearable sensors and software-based analysis, without requiring large-scale hardware production or hazardous materials.

The economic impact of this work is limited at the current stage. If developed further, a low-cost wearable cough monitoring system could reduce healthcare costs by enabling early detection and remote monitoring, but such claims require validation with larger and more diverse datasets.

6.3.2 Cost Analysis

The primary cost of this project is labor. Assuming the student researcher works approximately 10 hours per week during the semester at an hourly rate of \$15, the labor cost is

estimated to be \$2400 per semester and \$4800 for two semesters. Since the sensing hardware and data acquisition system are already available in the laboratory, no additional hardware costs are incurred.

6.3.3 Standards

If extended toward clinical use, the system should comply with relevant medical and software engineering standards. For medical device communication and interoperability, ISO/IEEE 11073 standards should be considered. The software development process should align with IEEE 12207 standards for software life cycle management.

In addition, compliance with Turkish and European Union medical device regulations (MDR) would be required to ensure patient safety, data privacy, and ethical use. Throughout the project, general engineering ethics and professional conduct principles are followed.

Appendix A

Supplementary Implementation Details

A.1 Model Training Setup and Hyperparameters

A.1.1 Cross-validation and data grouping

All window-level models were evaluated using **file-based GroupKFold** cross-validation to reduce data leakage between training and test sets. Each window inherits a `file_id`, and splits are created so that windows from the same file do not appear in both training and testing.

- Number of splits: **5** (`N_SPLITS = 5`)
- Random seed: **0** (`RANDOM_STATE = 0`)
- Output probabilities are collected as out-of-fold (OOF) predictions.

A.1.2 Class imbalance handling

Since cough windows are the minority class, balanced weighting is used:

- Logistic Regression and SVM: `class_weight="balanced"`
- Random Forest: `class_weight="balanced_subsample"`
- XGBoost: `scale_pos_weight =  $\frac{N_{neg}}{N_{pos}}$`

A.1.3 Model list and hyperparameters

Table A.1 summarizes the exact settings used in the notebook.

Table A.1: Window-level cough models and hyperparameters.

Model	Hyperparameters / Notes
LogReg_L2	Pipeline: StandardScaler + LogisticRegression; class_weight="balanced", max_iter=3000.
SVM_RBF	Pipeline: StandardScaler + SVC; kernel="rbf", class_weight="balanced", probability=True.
RF_300	RandomForestClassifier; n_estimators=300, class_weight="balanced_subsample", n_jobs=-1, random_state=0.
XGBoost	XGBClassifier; n_estimators=400, max_depth=4, learning_rate=0.05, scale_pos_weight as above, eval_metric="logloss", random_state=0.

A.2 Event-level Post-processing and IoU Matching Details

A.2.1 From window probabilities to cough events

Window-level probabilities are converted into cough event intervals using three steps:

1. **Thresholding:** windows with probability $\geq p_{\text{th}}$ are marked as positive.
2. **Grouping:** consecutive positive windows are merged into a candidate interval.
3. **Cleaning:** very short intervals are removed, and intervals separated by short gaps are merged.

Table A.2 lists the post-processing parameters used in the notebook.

Table A.2: Event post-processing parameters.

Parameter	Value
Probability threshold (p_{th})	0.95
Minimum event duration (T_{min})	0.15 s
Merge gap (T_{gap})	0.10 s
IoU threshold (τ_{IoU})	0.20

A.2.2 IoU definition

Predicted and ground-truth cough events are compared using Intersection-over-Union (IoU). For a predicted interval P and a ground-truth interval G ,

$$\text{IoU}(P, G) = \frac{|P \cap G|}{|P \cup G|}.$$

A predicted event is considered a match if its best IoU with any unused ground-truth event is at least τ_{IoU} .

A.2.3 Matching procedure

The matching is performed with a one-to-one assignment:

- For each predicted event, compute IoU with all ground-truth events.
- Select the ground-truth event with the highest IoU.
- If $\text{IoU} \geq \tau_{\text{IoU}}$ and the ground-truth event is not already matched, count as **TP**.
- Otherwise, count as **FP**.
- Any ground-truth event that is not matched is counted as **FN**.

A.2.4 Event-level metrics

Event-level precision, recall, and F1-score are computed as:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \quad \text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \quad \text{F1} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}.$$

Appendix B

Full Classification Reports (Window-level Cough)

This appendix reports the full window-level classification results for all models using out-of-fold predictions and a fixed decision threshold of 0.50. The evaluation is provided for two classes: *non-cough* and *cough*.

B.1 Random Forest (RF_300)

Table B.1: Window-level classification report for RF_300.

Class	Precision	Recall	F1-score	Support
non-cough	0.957	0.986	0.971	20128
cough	0.910	0.757	0.827	3692
Accuracy		0.951		23820
Macro avg	0.934	0.872	0.899	23820
Weighted avg	0.950	0.951	0.949	23820

B.2 XGBoost

Table B.2: Window-level classification report for XGBoost.

Class	Precision	Recall	F1-score	Support
non-cough	0.976	0.942	0.958	20128
cough	0.733	0.873	0.797	3692
Accuracy		0.931		23820
Macro avg	0.854	0.907	0.878	23820
Weighted avg	0.938	0.931	0.933	23820

B.3 SVM (RBF)

Table B.3: Window-level classification report for SVM_RBF.

Class	Precision	Recall	F1-score	Support
non-cough	0.959	0.972	0.965	20128
cough	0.836	0.771	0.802	3692
Accuracy		0.941		23820
Macro avg	0.897	0.872	0.884	23820
Weighted avg	0.940	0.941	0.940	23820

B.4 Logistic Regression (L2)

Table B.4: Window-level classification report for LogReg_L2.

Class	Precision	Recall	F1-score	Support
non-cough	0.972	0.870	0.918	20128
cough	0.549	0.865	0.672	3692
Accuracy		0.869		23820
Macro avg	0.761	0.867	0.795	23820
Weighted avg	0.907	0.869	0.880	23820

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